SECTION

EXPERIMENTAL DESIGNS

3

CHAPTER

EXPERIMENTAL DESIGNS, PART 1: INTRODUCTION

8

Every scientist has designed experiments. So what is there left for us to say about that topic that chemometrics/statistics can shed some light on? Well, quite a bit actually, since not all experiments are designed equally, but some are definitely more equal than others (to steal a paraphrase). Another way to say it is that every experiment is a designed experiment, but some designs are better than others (after all, nobody does experiments without SOME sort of a plan for it).

In point of fact, the sciences of both statistics and chemometrics have their own approach to how experiments should be designed, each with a view toward making experimental procedures "better" in some sense. There is a gradation between the two approaches; nevertheless, there is also somewhat of a distinction between what might be thought of as classical "statistical experimental design" and the more currently fashionable experimental designs considered from a chemometric point of view. These differences in approach reflect differences in the nature of the information to be obtained from each.

Experimental designs, and in particular "statistical" experimental designs, are used in order to achieve one or more of the following goals:

(1) Increase efficiency of resource use, that is, obtain the desired information using the fewest possible necessary experiments (this is usually what is thought of when "statistical experimental designs" are considered). This aspect of experimentation is particularly important when the experiment is large to begin with, or if the experiment uses resources that are rare or expensive, or if the experiment is destructive so that materials (especially expensive ones) are used up.

(2) Determine which variables or phenomena ("factors" in statistical/chemometric parlance) in an experiment are the "important" ones. This has two aspects: first, is an effect large enough that we can be sure it is real and not due simply to noise (or error) alone (i.e., "statistically significant.")? We have treated this question to some extent in our previous series of columns (SiS) and the eponymous book. The second aspect is if the effect of a factor is indeed real; is it of sufficiently large magnitude to be of practical importance? While the answer to this question is important to understanding the outcome of the experiment, it is not a statistical question, and we will give it fairly short shrift.

(3) Accommodate noise and/or other random error.

(4) Allow estimates to be made of the magnitude of the noise and/or other random error, if for no other reason than to compare our results to so as to tell if they are statistically significant.

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CHAPTER

9 EXPERIMENTAL DESIGNS, PART 2: ONE-WAY ANOVA☆

As we have mentioned in the previous chapter, "experimental design" often takes a form in scientific investigations, such that some of the experimental objects have been exposed to one level of the variable, while others have not been so exposed. Oftentimes this situation is called the "experimental subject" versus the "control subject" type of experiment. In the face of experimental error, or other source of variability of the readings, both the "experimental" and the "control" readings would be taken multiple times. That provides the information about the "natural" variability of the system against which the difference between the two can be compared. Then, a *t*-test is used to see if the difference between the "experimental" and "control" subjects is greater than can be accounted for by the inherent variability of the system. If it is, we conclude that the difference is "statistically significant" and that there is a real effect due to the "treatment" applied to the experimental subject. Of course there are variations on this theme: the difference between the "experimental" and "control" subjects can be due to different amounts of something applied to the two types of object, for example.

That is how we have treated this type of experiment previously. We will now consider a somewhat different way to formulate the same experiment: the purpose being able to set up the experimental design and the analysis of the data, in such a way that it can be generalized to more complicated types of experiments. In order to do this, we recognize that the value of any individual reading, whether from the experimental subject or the control subject, can be expressed as the sum of three quantities. These three quantities arise from a careful consideration of the nature of the data.

Given that a particular measurement belongs either to the experimental group or the control group, then the value of the data collected can be expressed as the sum of these three quantities:

1. The grand mean of all the data (experimental + control).
2. The difference between the mean of the data group (experimental or control) and the grand mean of the data.
3. The difference between the individual reading and the mean reading of its pertinent group.

This can then be expressed mathematically as:

$$X_{ij} = \overline{\overline{X}} + \left(\overline{X}_i - \overline{\overline{X}}\right) + \left(X_{ij} - \overline{X}_i\right) \tag{9-1}$$

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Table 9-6 Types of Experimental Designs

Number of Levels	Number of Factors	
	Single	**Multiple**
Two	Experimental versus control subjects	One-at-a-time designs Factorial designs Fractional factorial designs Nested designs Special designs
Multiple	Sensitivity testing Simple regression	Response surface designs Multiple regression

simple regression (often used in calibration work when only one physical variable in affected—in chemistry, electrochemical and chromatographic applications come to mind) (see Table 9-6 for examples).

The table is not exhaustive, although it does include a majority of experimental designs that are commonly used. One-at-a-time designs are the usual "nonstatistical" types of experiments that are often carried out by scientists in all disciplines. Not included explicitly, however, are experimental designs that are generated from combinations of listed items. For example, a multifactor experiment may have several levels of some of the factors but only two levels of other factors.

Also, due to the nature of the physical factors involved, the values of some of the factors may not be under the experimenter's control. Thus, some factors may be nested, while others may not be.

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[2] H. Mark, J. Workman, Statistics in spectroscopy - part 22 - and still counting, Spectroscopy 5 (3) (1990) 55–56.

CHAPTER

EXPERIMENTAL DESIGNS, PART 3: TWO-FACTOR DESIGNS☆ 10

As an example, consider a two-factor design with each factor at two levels. This is also a form of all-possible-combinations experiment. One item we note here is that there is more than one way to describe the form of an experiment, and we include a short digression here to explicate the multiplicity of ways of describing an experiment. In this particular case, we have two factors, each at two levels. One way we can describe it is as a listing of values corresponding to each experiment as shown in Table 10-1.

An alternative way we can describe it is as the experiment number that will correspond to each set of combinations of factors as in Table 10-2.

Whichever way we choose to describe these designs, they have some attractive features. We will illustrate these features with a numerical example. For our example, we will imagine an experiment where the scientist is interested in determining the influence of solvent and of catalyst on the yield of a chemical reaction. The questions to be answered are: does the solvent make a difference, and does the type of catalyst make a difference? The experiment is to consist of trying each of the four available catalysts and three solvents and determining the yield. The experiment can be described by Table 10-3.

In a more complicated experiment, where a physical variable such as temperature, which can be assigned meaningful physical values, where the physical variable and the sensitivity of the yield to temperature was of concern, we would then need to maintain the information regarding the actual temperatures.

For our first look at this experiment we will examine the behavior of a hypothetical experiment under two sets of conditions. The first set of conditions (scenario 1) produces the result under the following assumptions:

1. There is no influence of solvent.
2. There are no random influences on the experiment.
3. None of the catalysts have an effect.

The second set of conditions (scenario 2) is similar but with one change:

1. There is no influence of solvent.
2. There are no random influences on the experiment.
3. One of the catalysts has an effect.

In the second scenario, conditions 1 and 2 together mean that all results from the experiment will be the same as in the first scenario, and all results except the ones corresponding to the "effective" catalyst will

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CHAPTER

11 EXPERIMENTAL DESIGNS, PART 4: VARYING PARAMETERS TO EXPAND THE DESIGN☆

CHAPTER OUTLINE

Introduction to Factorial Designs 78
References 79

First, a clarification: We'd received some comments from a reader pointing out what appeared to be an error in the original Spectroscopy column [1] (corresponding to Chapter 15 in the first edition of this book and this chapter). In fact, there was no error as such, but there have been a couple of points of possible confusion in the article, which we take this opportunity to clear up. One of these corrections was actually done by a reader, and we refer all the rest of our readers to the letter to the editor of Spectroscopy, by Mohammad Yousaf [2], who raised the question of when an angle is "internal" versus "external" and then made the point that an "internal" angle is acute with respect to an axis and an "external" angle is obtuse with respect to an axis, when the angle is measured counterclockwise as is the usual procedure.

The other point of possible confusion was due to our inadvertent use of the same symbol (γ) to represent two different angles. We apologize for this and hope to set the record straight here. First, the angle γ shown in Fig. 1 of Ref. [3] (corresponding to Chapter 19 in this book and to Chapter 13 of the first edition) is not discussed.

The angle γ is calculated in Eq. (3) of Ref. [3] and Chapter 19. To quote from Chapter 19, "Because the third dimension is represented by the z-axis, we calculate the x-direction angle on the (x, z) plane as γ." The correct calculation for this angle is given in Eq. (19-3) of Chapter 19 of this book.

Thus, the angle α (the angle to the x-axis) and the angle β (the angle to the y-axis), as illustrated in the same figure, are both shown as a projection of the three-dimensional vector (2, 2, 6) onto the (x, y) plane, and the proper calculations for both α and β from what is then a two-dimensional vector are correct as given in Eqs. (1), (2) of that article [3] and Chapter 19. Again we apologize for the confusion.

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INTRODUCTION TO FACTORIAL DESIGNS

Back to our regularly scheduled discussion: We sometimes play ping-pong in these chapters, jumping back and forth from topic to topic as suits our fancy and hopefully keeps our readers interested (and on their toes—we seem to have succeeded at that): see the above (all feedback, good or bad, tells us that someone is at least reading this stuff!). So it is with our current chapter. We have discussed experimental designs in the columns corresponding to Chapters 8–10, taken a break, and are now returning to the topic. In those previous chapters, the designs we discussed were, with the exception of one particularly nifty design (that represents a special case of a more general type of design that we will eventually get around to discussing), rather simple and plain, in the sense that the designs included only small numbers of levels of the various factors of interest and were basically considerations of "all possible combinations" of those factors—the types of experiments that scientists have been designing "forever" without any thought or consideration that they were "statistical experimental designs." Obviously, though, since they represent special cases of wider classes of designs, they must also come under that umbrella. So what is different about the experimental designs that we call "statistical" or "chemometric" designs? Actually, very little, until we take a look at what happens when we need to scale these designs up to larger sample numbers or more complex designs.

Before we do that, let us consider the various types of experiments, and the nature of the factors that are used in those experiments, involved. Someone doing an experiment is generally trying to learn about the effect of some phenomenon on some quantity that can be measured. While there are cases that don't fit the description we're about to present, one very common type of experiment involves changing (or allowing to change) some parameter and then measuring the effect of that change. If there is only one such parameter, the situation is pretty straightforward, but things start getting interesting when two or more possible parameters are involved. Intuitively, one's first instinct is to measure the results that are obtained for all possible combinations of the available values of the parameters.

In one of our columns of our previous series [4], reprinted in Ref. [5], we looked at some experiments that involved two parameters (factors), each at two levels. In our previous book [5], we also briefly looked at a three-factor, two-level design, with attention to how it could be represented geometrically. The use of the term "three factor, two level" to describe the design means that each factor was present at two levels, that is, the corresponding parameters were each permitted to assume two values.

There are several ways we can expand a design such as this: we can increase the number of factors, the number of levels of each factor, or, of course, we can do both. There are other differences that can be superimposed over the basic idea of the simple, all-possible combinations of factors, such as to consider the effect of whether we can control the levels of the factors (if we can, then we can do things that are not possible to do when we cannot control the levels of the factors), whether the "levels" correspond to physical characteristics that can be evaluated, and whether the values described have real physical meaning (e.g., temperature has real physical meaning, while catalyst type does not, even though different catalysts in an experiment may all have different degrees of effectiveness, and reproducibly so).

Another consideration is whether all the factors can be changed independently through their range of possible values or whether there are limits on the possible values. The most obvious limiting situation is the case of mixtures, where all the components of a mixture must sum to 100%. Other limitations might be imposed by the physical (or chemical) constraints on the behavior of the materials

involved: solubility as a function of temperature, for example, or as a function of other materials present (e.g., maximum solubility of salt in water-alcohol mixtures will vary with the ratio of the two solvents). Other limits might be set by practical considerations such as safety. Except for specialized work by scientists experienced in the field, few experimenters would want to work, for example, with materials at concentrations above their explosive limits.

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CHAPTER

12

EXPERIMENTAL DESIGNS, PART 5: ONE-AT-A-TIME DESIGNS☆

In the previous chapter we began our discussions of factorial designs. If we expand the basic n-factor two-level experiment by increasing the number of factors, maintaining the restriction of allowing each to assume only 2 values, then the number of experiments required is 2^n, where n is the number of factors. Even for experiments that are easy to perform, this number quickly gets out of hand; if 8 different factors are of interest, the number of experiments needed to determine the effect of all possible combinations is 256, and this number increases exponentially.

The other obvious way we might want to expand the experiment is to increase the number of levels (values) that some or all of the factors take. In this case, the number of experiments required increases even faster than 2^n. So, for example, if each factor is at three levels then the number of experiments needed is 3^n (for eight factors, corresponding to our previous calculation, this comes to 6561 experiments for all combinations at all levels!). In the general case, the number of experiments needed is $\Pi_i n_i$, where n_i is the number of levels of the ith factor.

It should be clear at this point that the problem with this scenario is the sheer number of experiments needed, which in the real world translates into time, resources, and expense. Something must be done.

Over the years, several "somethings" have been done. The intuitive experimenter, expert in his particular field of science but untrained in "statistical" designs, simplifies the whole process by throwing out all the combinations and uses what are known as simply "one-at-a-time" designs [1]. Daniels describes five variations of this basic design, but basically these are only useful when the random noise or error is small (compared to the expected magnitude of the effects) and involve the experimenter changing one variable (factor) at a time to see which one(s) cause the greatest effect. Sometimes those are then examined in greater detail, by varying them over larger ranges, and/or at more values lying within the original range. This solves the problem of the proliferation of experiments, since the number of experiments needed is now only $1+\Sigma_i n_i$ a much smaller number than $\Pi_i n_i$. It also provides a first-order indication of the effect of each of the factors. The difficulty with this approach is the possibility of throwing out the baby with the bathwater, so to speak, by losing all information about the actual noise level, as well as any information about any possible synergistic or inhibitory interactions between the factors.

Thus, when statisticians got into the act, they saw the need to retain the information that was not included in the one-at-a-time plans, while still keeping the total number of experiments manageable, giving rise to the birth of "statistical experimental designs." Several types of "statistical experimental

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designs" have been developed over the years, with, of course, innumerable variations. However, they can be placed into a fairly small group of main design types:

1. Factorial
2. Fractional factorial
3. Sequential
4. (a) Latin square
   (b) Graeco-Latin square
   (c) Latin and Graeco-Latin cubes
5. Model building
6. Response surface

By far the most statistical energy has been spent on the design and analysis of factorial designs. Books dealing with such designs (e.g., see Refs. [2,3]) spend a good part of their space discussing the variations required to accommodate such considerations as replication, blocking, how to deal with situations where the experiment itself is destructive (so that the same specimen is never available for retesting), whether the experimental conditions can be reproduced at will, and whether or not the experimental factors (or the desired response) can be assigned meaningful numerical values. Each of these considerations dictates the types of designs that can be considered and how they must be implemented.

For our current discussions, however, we have been taking the path of discussing ways to reduce the required number of experiments, while still retaining the advantages of obtaining several types of information about the system under consideration. The simplest such type of design is the sequential design, simplest if for no other reason than that the type of design it replaces is one of the simplest designs itself. We will discuss this type of design in the next chapter.

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CHAPTER

13 EXPERIMENTAL DESIGNS, PART 6: SEQUENTIAL DESIGNS☆

The simplest such type of resource-conserving design is the sequential design, simplest if for no other reason than that the type of design it replaces, the simple sample-versus-control design, is one of the simplest designs itself. This design is the simple test for comparison of means, using the *Z*-test or the *t*-test as the test statistic; we have discussed these in our previous book: "Statistics in Spectroscopy" [1].

The standard *t*-test (or *Z*-test, if conditions permit) specifies a predefined number of measurements to be made, either for a single condition or for a pair of conditions (i.e., sample vs "control"). The difference between the two states is compared to the experimental error evidenced in the data and a decision made based on whether the difference between the sample behavior at the different conditions is "large enough": compared to the noise (or error) to warrant the conclusion that there is a (statistically significant) difference between them.

For a sequential test, the number of experiments is not predefined. Rather, experiments are performed sequentially (surprise!), and the series terminated as soon as enough data is available that a decision can be made as to whether the difference is "large enough." True, it is theoretically possible for such a sequence of experiments to be indefinitely long; in practice, however, it is far more common for the situation to become decidable after fewer experiments than are required for the case of a fixed number of experiments.

So how does this "magic" experimental design work? The best available discussion we know of is in Ref. [2]. The standard concept behind this experimental design is illustrated in Fig. 13-1. As this figure shows, the "universe" is divided into three regions: the region (A) is the region of acceptance of the null hypothesis and region C is the region of acceptance of the alternative hypothesis (or rejection of the null hypothesis). The middle region, B, is the region of continuation: as long as values fall into this region, it means there is not enough information to make a decision; therefore, we must continue with the experiments.

Fig. 13-2 shows how this works for two (hypothetical) typical cases. First a single experiment is performed, and the results are noted. If these results are put it into the region of continuing the project (virtually inevitable after only one experiment), then a second experiment is performed and so forth. Fig. 13-2 shows typical results for two possible sequences of experiments: the one indicated by the circles enters the region of acceptance of the alternative hypothesis after nine experiments, the one indicated by the crosses enters the region of acceptance of the null hypothesis after seven experiments. Obviously, the actual number of experiments required will depend on both the nature of the

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While a discussion of β properly belongs in the realm of elementary statistics, at this point, it is worthwhile to go back to some of those discussions, to examine how this impacts our current interests. We will proceed along with this digression in our next chapter.

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CHAPTER

14

EXPERIMENTAL DESIGNS, PART 7: β, THE POWER OF A TEST☆

In the previous chapter, we started discussing the way a series of experiments could be designed so that the decision to perform another experiment could be based on the outcomes of the experiments already done. We saw there that we needed to be able to tell if we could stop because the result had become statistically significant, and we also saw that we needed a way to tell if we could stop because we had reached the statistically significant conclusion that there is no real difference between the sample and the (hypothetical) reference population. This is necessary, indeed crucial; otherwise, we could continue experimenting endlessly, waiting for a statistically significant result when there was no real difference to detect so that none would be expected.

The first stopping criterion is straightforward, it is simply the standard hypothesis test, based on α probabilities that we have previously discussed, of a sample coming from the hypothesized population P_0 [1]. The second stopping criterion, however, seems to fly in the face of our previous discussions on the topic, where we said that you could not *prove* two populations the same.

However, it is easier to see the reason for the second statement, that there is difficulty in proving that a sample came from a given population, if we reword the statement of it. We do that by making it a double negative and ask ourselves, can we prove that it did not come from a different population? Now, the nature of the difficulty becomes clearer: we have no information about the nature of the "different" population that we want to test against.

Now that we can see the problem, we can find a point of attack against it. We can hypothesize an alternative population (P_a) with any given characteristics we want and then consider the consequences of dealing with that alternate population. In particular, we consider the probabilities of either accepting or rejecting our original null hypothesis (based on P_0) if, in fact our sample came from the alternate population P_a. The probability of coming to the incorrect conclusion, that the sample came from P_0 when it really came from P_a, is called the β probability (compare with α probability, which is the probability of drawing the incorrect conclusion that a sample did not come from P_0 when it really did). This is known in statistical parlance as the "power" of the statistical test.

Thus, in performing a statistical hypothesis test, we would normally consider only the ordinary tests against the alpha error as a means of determining statistical significance. However, as we have seen, that leaves completely open the number of samples needed. The power of a test gives us a criterion which will allow determining the number of samples. To redefine the term: the power of a statistical test is the probability of obtaining a statistically significant result *given that in fact the null hypothesis is false*. Ordinarily, to achieve a nonsignificant result is easy: just use few enough samples. To show that

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depends on which population it came from, that it would fall above (or below) the upper critical limit indicated for H_0.

The shaded areas in Fig. 14-1 indicate the probabilities for a random sample falling below the critical value for H_0, when in fact one of those alternate populations is in fact the correct population from which the sample was taken. As can be seen, these probabilities are roughly 50% for population P_1 and roughly 5% for population P_2. These probabilities are the probabilities of (incorrectly) concluding that the data is consistent with H_0, for these two hypothetical alternative populations.

(continued in our next chapter)

REFERENCE

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CHAPTER

15

EXPERIMENTAL DESIGNS, PART 8: β, THE POWER OF A TEST (CONTINUED)☆

Continuing from our previous discussion in Chapter 14, analogous to making what we have called (and is the standard statistical terminology) the α error when the data is above the critical value but is really from P_0, this new error is called the β error, and the corresponding probability is called the β probability. As a caveat, we must note that the correct value of β can be obtained only subject to the usual considerations of all statistical calculations: errors are random and independent and so on. In addition, since we don't really know the characteristics of the alternate population, we must make additional assumptions. One of these assumptions is that the errors are Normally distributed (i.e., follow a Gaussian distribution) and that the standard deviation of the alternate population (P_a) is the same as that of the hypothesized population (P_0), regardless of the value of its mean.

The existence of the β probability provides us with the tool for determining what is called the *power* of the test, which is just $1-\beta$, the probability of coming to the correct conclusion when in fact the data did not come from the hypothesized population P_0. This is the answer to our earlier question: once we have defined the alternate population P_a, we can determine the β probability of a sample having come from P_a, just as we can determine the α probability of that sample having come from P_0.

So how does this help us determine n? As we know from our previous discussion of the Central Limit Theorem [1], the standard deviation of the average of a sample from a given population decreases from the population standard deviation monotonically as n increases. Thus, we can fix μ_0 and μ_a and adjust the α and β probabilities by adjusting n and the critical value.

Normally, it is convenient to adjust the critical value to be equidistant from μ_0 and μ_a and then adjust n so that that critical value represents the desired probability levels for α and β.

As an example, we can set alpha and beta levels to the same value, which makes for a simple computation of the number of samples needed, at least for the simple case we have been considering: the comparison of means. If we use the 95% value for both (a very stringent test), which corresponds to a Z-value of 1.96 (as we know), then if we let D represent the difference in means between the two values (sample data and population mean), and S is the precision of the data, we find that

$$D \geq 3.92S/\sqrt{n} \tag{15-1}$$

so that

$$n = (3.92S/D)^2 = 15(S/D)^2 \tag{15-2}$$

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In words, we would need 15 samples for 95% confidence on both alpha and beta, to distinguish a difference of the means equal to the precision of the measurement, and the number of samples increases as the square of any decreases in the difference we want to detect.

To compute the power for a hypothesis test based on standard deviation, we would have to read off the corresponding probability points from a chi-square table; for 95% confidences on both alpha and beta, the square root of the ratio of $\chi^2(0.95, v)$ and $\chi^2(0.05, v)$ ($v =$ the degrees of freedom, close enough to n for now) is the ratio of standard deviations that can be distinguished at that level of power. Similarly, to the case of the means, v would also be related to the square of that ratio, but χ^2 would still have to be read from tables (or computed numerically). As an example, for 35 samples, the precision of the instrument could not be tested to be better than:

$$\sqrt{48.6/21.6} = \sqrt{2.25} = 1.5 \tag{15-3}$$

or 1.5 times the precision of the reference method with that amount of power, and as before, the required value for n will increase as the square of any improvement we want to demonstrate. The ratio of $\chi^2(0.95, v)$ to $\chi^2(0.05, v)$ does decrease as v increases but not nearly as fast as the square increases: to improve the results this way is a losing fight.

Thus, the use of the concept of the power of a test allows specification of the number of samples (although it may turn out to be very high) and by virtue of that, forms the basis for performing experiments as a sequential series, as described in Chapter 13.

REFERENCE

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CHAPTER

16

EXPERIMENTAL DESIGNS, PART 9: SEQUENTIAL DESIGNS (CONCLUDED)☆

Chapters 14 and 15 describe how the use of the power concept for a hypothesis test allows us to determine a value for n at which we can state with both $\alpha\%$ and $\beta\%$ certainty that the given data either is or is not consistent with the stated null hypothesis H_0. To recap those results briefly, as a lead-in for returning to our main topic (from Chapter 13), we showed that the concept of the power of a statistical hypothesis test allowed us to determine both the α and β probabilities and that these two knowns allowed us to then determine, for every n, what was otherwise a "floating" (or basically, an unknown) quantity, D.

At this point it should start to become clear what is going on. If a given set of α, β, and D allow us to determine n, then similarly, a corresponding set of α, β, and n allow us to determine D. Thus, for a given α and β, n and D are functions of each other, and it then becomes a simple matter (at least in principle, in practice the math involved is extremely hairy) to determine the functionality.

In fact the actual situation is considerably more complicated to determine mathematically. In our previous discussions, we have made a number of simplifying assumptions which cannot be used if we wish to calculate correct values for our expressions and for which the actual situation must be incorporated into the math.

The first of these assumptions is the use of the normal distribution. When we perform an experiment using a sequential design, we are implicitly using the experimentally determined value of s, the sample SD, against which to compare the difference between the data and the hypothesis. As we have discussed previously, the use of the experimental value of s for the SD, rather than the population value of σ, means that we must use the t-distribution as the basis of our comparisons, rather than the normal distribution. This, of course, causes a change in the critical value we must consider, especially at small values of n (which is where we want to be working, after all, since the purpose of a sequential test is to minimize the required number of experiments).

The other key assumption that we sort of implied was that the comparison SD is constant. Of course we know that as n changes, the comparison value changes as the square root of n. This is on top of and in addition to the changes caused by the use of the t rather than the normal (Z) distribution.

So how is this related to the nature of the graph used for the sequential experimental design? We forgo the detailed math here, in deference to trying to impart an intuitive grasp of the topic, and in Chapter 15 we have already presented the equations involved.

The limits of the allowable values around the hypothesized values close in on it as n increases. This behavior is shown in Fig. 16-1. If, in fact, we were to plot the mean of the population as a function of n,

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